

ALAKAZAM

The in-battle coach — a plain brief

The final, hardest piece of ABRA, a system that learns competitive Pokémon from thousands of real games.

What it is

A coach that, at any moment in a battle, tells you the best move to make and how likely you are to win.

How it plays

Like a strong human thinking hard: it keeps a running guess of what the opponent is hiding, pictures how the next few turns could unfold, and picks the move that wins most often. Two things make it strong — it **plans ahead** instead of just reacting, and it keeps its choices **unpredictable** so it can't be read.

What it runs on

Using it is light — it runs in a web browser on an ordinary laptop, no special hardware. Building the strongest version is the heavy part: it must learn from a huge pile of past matches and from playing against itself, which needs a powerful rented cloud computer (a GPU) for a while.

How long

A basic but genuinely useful version: about **one to two weeks**. A version that could beat top human players: **a few months** — and most of that time is spent letting it play itself to learn, not the training itself.

The honest limit

Some things simply can't be predicted — like who wins from the starting line-ups alone, which is close to a coin flip. So ALAKAZAM is judged on making **good decisions in the moment**, not on calling the winner in advance.

A few terms

Imperfect information — you can't see everything about the opponent's position, unlike chess.

Simultaneous move — both players lock in a move at once and reveal together, like rock-paper-scissors.

Mixed strategy — choosing randomly among options with fixed odds, so you can't be predicted and countered.

Belief — the AI's running probability estimate of what the opponent is hiding, updated as clues appear.

Self-play — the AI plays millions of games against copies of itself to make its own training data.

GPU — a powerful graphics chip used to train large AI models, far faster at it than a laptop.